

1. Consider the discrete-time signal $\mathbf{X} = \{1, 2, 3, 4, 5\}$ and the linear filter with impulse response $\mathbf{a} = \{\frac{1}{2}, 1, \frac{1}{2}\}$.
 - (a) Compute the filtered series $\mathbf{Y}$ using zero-padding for values outside the range of the signal.

Solution: We compute the filtered output Y_t for each value of t .

For $t = 0$:

$$Y_0 = a_0X_0 + a_1X_{-1} + a_2X_{-2}.$$

Since $X_{-1} = 0$ and $X_{-2} = 0$ (due to zero-padding), we get:

$$Y_0 = \frac{1}{2} \cdot 1 + 1 \cdot 0 + \frac{1}{2} \cdot 0 = \frac{1}{2}.$$

For $t = 1$:

$$Y_1 = a_0X_1 + a_1X_0 + a_2X_{-1} = \frac{1}{2} \cdot 2 + 1 \cdot 1 + \frac{1}{2} \cdot 0 = 1 + 1 = 2.$$

For $t = 2$:

$$Y_2 = a_0X_2 + a_1X_1 + a_2X_0 = \frac{1}{2} \cdot 3 + 1 \cdot 2 + \frac{1}{2} \cdot 1 = \frac{3}{2} + 2 + \frac{1}{2} = 4.$$

For $t = 3$:

$$Y_3 = a_0X_3 + a_1X_2 + a_2X_1 = \frac{1}{2} \cdot 4 + 1 \cdot 3 + \frac{1}{2} \cdot 2 = 2 + 3 + 1 = 6.$$

For $t = 4$:

$$Y_4 = a_0X_4 + a_1X_3 + a_2X_2 = \frac{1}{2} \cdot 5 + 1 \cdot 4 + \frac{1}{2} \cdot 3 = \frac{5}{2} + 4 + \frac{3}{2} = 8.$$

Thus, the filtered series is:

$$\mathbf{Y} = \left\{ \frac{1}{2}, 2, 4, 6, 8 \right\}.$$

- (b) Compute the frequency response of the filter.

Solution: The frequency response A of a discrete-time linear filter is the Fourier transform of its impulse response a_j .

$$A(\omega_j) = \frac{1}{n} \sum_{t=0}^{n-1} a_t e^{-2\pi i \omega_j t}.$$

Considering $n = 5$ (the size of X) and substituting the values of the impulse response with zero-padding:

$$A(\omega_j) = \frac{1}{5} \left[\frac{1}{2} e^{-2\pi i \omega_j 0} + 1 e^{-2\pi i \omega_j 1} + \frac{1}{2} e^{-2\pi i \omega_j 2} + 0 e^{-2\pi i \omega_j 3} + 0 e^{-2\pi i \omega_j 4} \right].$$

This simplifies to:

$$A(\omega_j) = \frac{1}{5} \left(\frac{1}{2} + e^{-2\pi i \omega_j} + \frac{1}{2} e^{-4\pi i \omega_j} \right).$$

2. Consider a moving average smoothing filter of window size $L = 2m + 1$, where m is an integer.

- (a) Write the impulse response $a(t)$.

Solution: The impulse response of the moving average filter is defined as the weights applied to the signal in the time domain. For a window size $L = 2m + 1$, the filter applies equal weights to L contiguous values of the signal. Therefore, the impulse response is given by:

$$a(t) = \begin{cases} \frac{1}{L}, & \text{for } |t| \leq m, \\ 0, & \text{otherwise.} \end{cases}$$

- (b) Compute the frequency response $A(\omega)$.

Solution: The frequency response $A(\omega)$ of the filter is the Fourier transform of the impulse response $a(t)$. For the moving average filter, the Fourier transform is given by:

$$A(\omega) = \sum_{t=0}^{L-1} \frac{1}{L} e^{-i\omega t} = \frac{1}{L} \sum_{t=0}^{L-1} (e^{-i\omega})^t.$$

This is a geometric series, the sum results in:

$$A(\omega) = \frac{1}{L} \cdot \frac{1 - e^{-i\omega L}}{1 - e^{-i\omega}}.$$

- (c) Compute the power response.

Solution: The power response is the magnitude squared of the frequency response:

$$|A(\omega)|^2 = \left| \frac{1}{L} \cdot \frac{1 - e^{-i\omega L}}{1 - e^{-i\omega}} \right|^2 = \frac{1}{L^2} \frac{|1 - e^{-i\omega L}|^2}{|1 - e^{-i\omega}|^2}.$$

Expanding the squares, we get

$$\begin{aligned} |A(\omega)|^2 &= \frac{1}{L^2} \frac{(1 - \cos(\omega L))^2 + \sin^2(\omega L)}{(1 - \cos(\omega))^2 + \sin^2(\omega)} \\ &= \frac{1}{L^2} \frac{1 - 2\cos(\omega L) + \cos^2(\omega L) + \sin^2(\omega L)}{1 - 2\cos(\omega) + \cos^2(\omega) + \sin^2(\omega)} \\ &= \frac{1}{L^2} \frac{1 - \cos(\omega L)}{1 - \cos(\omega)} \end{aligned}$$

Since $1 - \cos(2x) = \sin^2(x)$, we can rewrite the equation as

$$|A(\omega)|^2 = \frac{1}{L^2} \frac{\sin^2(\omega L/2)}{\sin^2(\omega/2)}$$

Hence, the power response is the squared rescaled Dirichlet kernel.

- (d) What are the disadvantages of using this filter as low-pass filter?

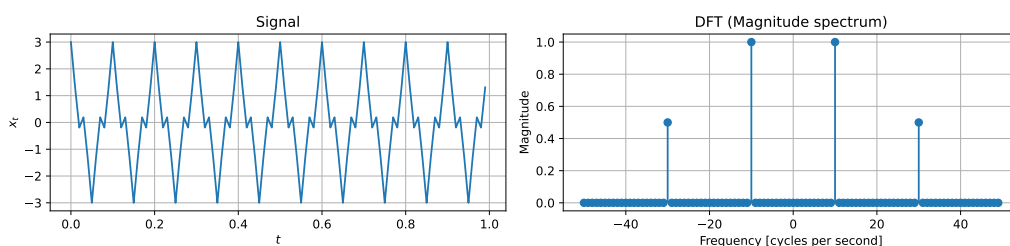
Solution: Although the moving average filter can act as a simple low-pass filter, it has several disadvantages:

- (a) Poor frequency selectivity: The moving average filter has a broad transition band between the passband and stopband. This means that frequencies outside the passband are not sharply attenuated, leading to a less precise frequency cutoff.
- (b) Ring effects: The impulse response of a moving average filter is not causal in the sense that it uses future values of the signal. This results in "ringing" or overshoot near the cutoff frequency when the filter is applied.
- (c) Limited performance at high frequencies: The moving average filter does not fully remove high frequencies but distort them.
- (d) Edge effects: When applied to real signals, especially those with sharp transitions (such as edges in image processing), the moving average filter can blur the edges, making it less effective for tasks like edge detection.

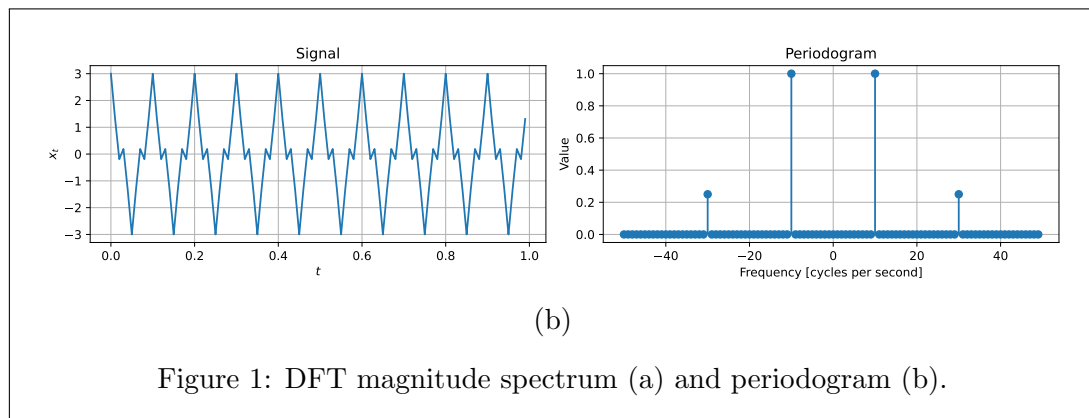
3. Consider the signal $x_t = 2 \cos(2\pi \cdot 10t) + \cos(2\pi \cdot 30t)$ sampled at 100 Hz for 1 second.

- (a) Sketch the DFT magnitude spectrum and the periodogram of the signal, including both positive and negative frequencies.

Solution:



(a)



(b)

Figure 1: DFT magnitude spectrum (a) and periodogram (b).

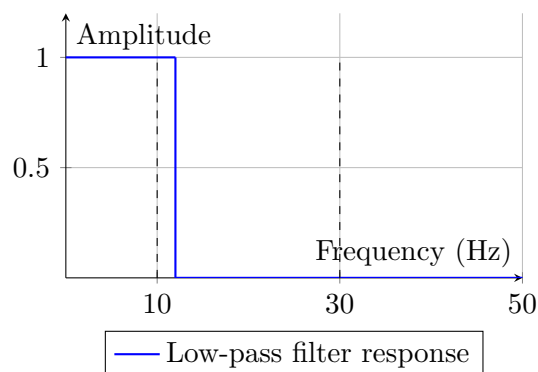
- (b) Imagine you only want to retain the 10 Hz component, what type of linear filter would you use? Sketch the frequency response of the filter.

Solution: To retain only the 10 Hz component from the signal, we would use a **low-pass filter**. A low-pass filter allows frequencies below a certain cutoff frequency to pass through while attenuating higher frequencies.

Filter Design:

- The cutoff frequency should be slightly above 10 Hz (e.g., 12 Hz) to ensure the 10 Hz component is retained without distortion.
- The filter should strongly attenuate frequencies higher than 12 Hz, particularly around 30 Hz, to remove the unwanted component.
- For practical implementation, we could design a low-pass filter using techniques such as a Butterworth filter or a finite impulse response (FIR) filter.

Frequency Response Sketch: A sketch of the idealized frequency response of the low-pass filter is shown below:



Explanation of the Frequency Response:

- Frequencies below the cutoff (12 Hz) are passed with a gain of 1.
- Frequencies above the cutoff are attenuated to zero (ideal case).
- The 10 Hz component is retained, while the 30 Hz component is removed.

- (c) Now, consider the signal shown in fig. 2, along its discrete Fourier transform (DFT)

magnitude spectrum. The signal is sampled at 800 Hz, and no aliasing is present.

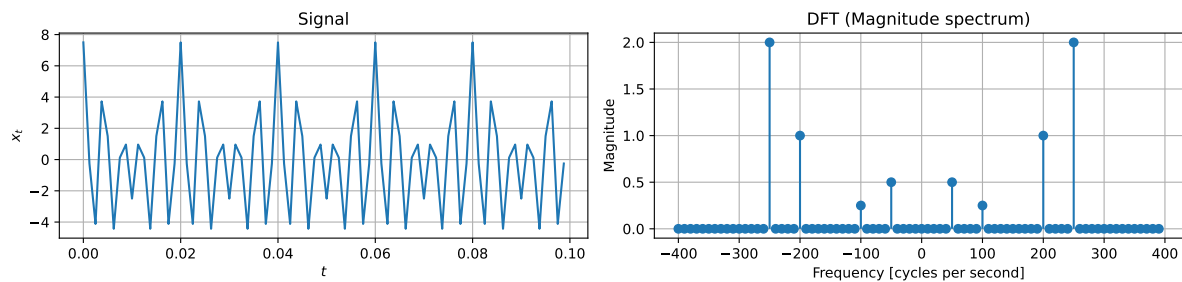


Figure 2: Signal and its DFT magnitude spectrum.

Write the mathematical representation of the signal in terms of its frequency components.

Solution: From the DFT we can deduce that the signal x_t is a sum of cosine waves at four distinct frequencies, each with a different amplitude. Based on the figure, the frequencies and amplitudes are:

- Frequency: 50 Hz, Amplitude: 1,
- Frequency: 100 Hz, Amplitude: 0.5,
- Frequency: 200 Hz, Amplitude: 2,
- Frequency: 250 Hz, Amplitude: 4.

The mathematical representation of the signal is:

$$x_t = \cos(2\pi \cdot 50 \cdot t) + 0.5 \cos(2\pi \cdot 100 \cdot t) + 2 \cos(2\pi \cdot 200 \cdot t) + 4 \cos(2\pi \cdot 250 \cdot t),$$

where t is the time variable, and the signal is sampled at a rate of 800 Hz.

- (d) Suppose the signal in fig. 2 is sampled at a rate of 450 Hz. Sketch the updated DFT magnitude spectrum and explain the changes. Determine the minimum sampling frequency required to avoid aliasing for this signal.

Solution: Effect of Sampling at 450 Hz:

The signal x_t contains frequency components at 50 Hz, 100 Hz, 200 Hz, and 250 Hz. When the signal is sampled at 450 Hz, any frequencies exceeding the Nyquist frequency (half the sampling rate, $f_s/2 = 225$ Hz) will alias into the range $[0, 225]$ Hz.

Aliasing:

- The 50 Hz, 100 Hz, and 200 Hz components are below the Nyquist frequency and remain unaffected.
- The 250 Hz component exceeds the Nyquist frequency and aliases. The aliased frequency can be computed as:

$$f_{\text{aliased}} = |f_{\text{original}} - kf_s|, \quad k = \text{integer such that } f_{\text{aliased}} \in [0, f_s/2].$$

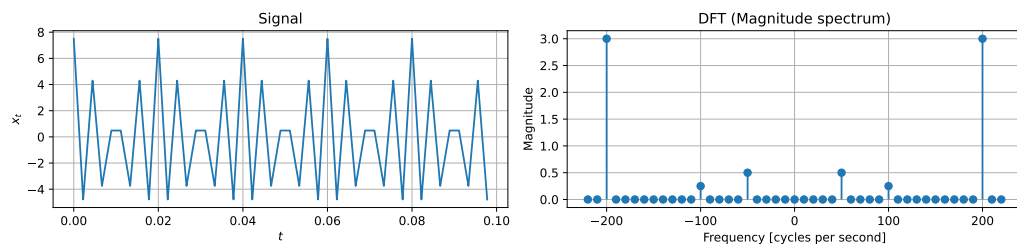
For $f_{\text{original}} = 250$ Hz:

$$f_{\text{aliased}} = |250 - 450| = 200 \text{ Hz.}$$

This means the 250 Hz component aliases into the 200 Hz component.

Therefore, the updated DFT spectrum will show peaks at:

- 50 Hz (amplitude 1),
- 100 Hz (amplitude 0.5),
- 200 Hz (amplitude $2 + 4 = 6$) due to the aliasing of the 250 Hz component.



Minimum Sampling Frequency to Avoid Aliasing:

The highest frequency component in the signal is 250 Hz. To avoid aliasing, the sampling frequency must satisfy the Nyquist criterion. Thus, the minimum sampling frequency required to avoid aliasing is 500 Hz.